

**ATTACHMENT 2 – CONTRACTOR TRANSMITTAL (SUBMITTAL NUMBER
723.480-014 C) FROM BECHTEL / PARSONS BRINCKERHOFF (AUTHORIZED
REPRESENTATIVE) TO MODERN CONTINENTAL COMPANY DATED
JANUARY 7, 2000 REGARDING THE CONTRACTOR’S ANCHOR ADEQUACY
SUBMITTAL CERTIFIED BY PROFESSIONAL ENGINEER (PE) SEAL**

(34 pages)



Central Artery/Tunnel

CONTRACTOR TRANSMITTAL

TRANSMITTAL DATE:

12/30/97

TRANSMITTAL NO:

05-0344

CONTRACTOR:

Modern Construction

ATTENTION:

Robert Steffy

CONTRACT NO:

9919-C0912

DOCUMENT TYPE

- R - REPRODUCIBLE
- P - PRINT
- C - CATALOG CUT
- E - EQUIPMENT MANUAL
- S - SAMPLE
- T - TECHNICAL DATA OR CALCULATIONS

REVIEW STATUS

- 1 - WORK MAY PROCEED.
- 2M - WORK MAY PROCEED SUBJECT TO THE CHANGES INDICATED. RESUBMITTAL NOT REQUIRED.
- 2R - WORK MAY NOT PROCEED ON INDICATED PART(S). RESUBMITTAL REQUIRED. WORK MAY PROCEED ON OTHER PART(S).
- 3 - WORK MAY NOT PROCEED. REVISE AND RESUBMIT.
- 4 - FOR INFORMATION ONLY.

DATE RECEIVED BY BIPB RE OFFICE:

12-30-97

DATE DUE BACK TO CONTRACTOR:

1-27-00

DATE SENT BACK TO CONTRACTOR:

1-7-00

RESIDENT ENGINEER:

DATE:

1/7/00

ITEM	SUBMITTAL NO. (I.E., 120.010-011-A)			DESCRIPTION	DOC DATE	DOC TYPE	REVIEW STATUS
	SPEC SECT	- SEQ#	- REV				
1	123-480	014	C	STRUCTURAL CALCULATIONS FOR ADHESIVE ANCHORS		T	ZN
2							
3							
4							
5							
6							
7							

COMMENTS TO CONTRACTOR (USE ADDITIONAL SHEETS AS NECESSARY)

Provide final report # ER 4514 when

Issued.

REVIEWER:

DATE:

1-7-00

WHITE - REVIEWER'S COPY GREEN - RESIDENT ENGINEER'S COPY YELLOW - INFORMATION COPY PINK - PROJECT ENGINEER COPY GOLD - CONTRACTOR'S COPY

CR051 (01/96)



Structural Calculations for Adhesive Anchors

INDEX STAMP

Contract No. 99119-C09B2	Paragraph 1.04 K
Specification Section (Spec-Seq-Rev)	
723.480-014 C	

CENTRAL ARTERY / TUNNEL

Permission To Proceed Does Not Constitute Acceptance Or Approval Of Design Details, Calculations, Analysis, Test Methods, Or Materials Developed Or Selected By The Contractor / Supplier And Does Not Relieve The Contractor / Supplier From Full Compliance With Contract Requirements.

☐	1. Work May Proceed
☒	2N. Work May Proceed Subject To The Changes Indicated Resubmittal Not Required
☐	2R. Work May Not Proceed On Indicated Parts Resubmittal Required. Work May Proceed on Other Parts.
☐	3. Revise and Resubmit. Work May Not Proceed.
☐	4. For Information Only.

REVIEWER	
Signature	
Company	
Date	1-7-00

K:\9809\SUBMIT\SUBSTAMP.WK4

Section 1

Manufacturer's Response to Comments from Previous Submittal

Powers FASTENERS

Monday, December 20, 1999

Newman, Renner, Colony
19 South Maple Street
Bellingham, MA 02019

Attn: Mr. Gary Wilcox
Ph: (508) 966-4455
Fax: (508) 966-5255

Powers Fasteners, Inc.
2 Powers Square
New Rochelle, NY
USA 10801

Telephone: 1-(914) 238-8300
Fax: 1-(914) 878-8483

Web Site: www.powers.com
E-Mail: williamson@powers.com

Re: Modern Continental - CO9B2 Tunnel Finishes - Epoxy Anchor

Dear Mr. Wilcox:

Pursuant to your request, the attachments forwarded under your fax cover sheets dated December 6 and 15, 1999 have been reviewed. The following comments are in response to hand written inquiries in order of their appearance:

Document Title: CO9B2 Structural Calculations (ref. Submittal 723.480-014C)
Page Number 1

Section B: "Allowable Loads", Adhesive allowables

a. Adhesive Tension Allowable = 6,350 lbs.

Powers Fasteners' comment: Agree - Adhesive Tension Allowable = 6,350 lbs.

Section C: "Interaction Factors" $\{(T_{actual} / T_{allowable}) + (V_{actual} / V_{allowable})\} < 1$

Powers Fasteners' comment: The versions of this equation that appear in current ICBO Acceptance Criteria for Adhesive Anchors in Concrete and Masonry Elements AC508 are attached. When final issuance of ICBO ES, Inc. Evaluation Report No. 4514 which includes Power-Fast becomes available, the equation that will appear in the report will be

$(P_s / P_t) + (V_s / V_t) \leq 1$ where:

P_s = Applied service tension load

P_t = Service tension load

V_s = Applied service shear load

V_t = Service shear load

A draft copy of ICBO ES, Inc. Evaluation Report No. 4514 is attached.

Page Number 2

Section C: "Interaction Factors", 2. Two bolt connection loading condition (F.S. = 2.0)

1) Powers Fasteners' comments: Please refer to page 143 of Powers' Fastening Systems Design Manual, Second Edition, Section 51.3 Performance Data. It is stated: "... (adhesive load) capacities listed in this manual are published in two ways. The ultimate or failure load for the adhesive is published first. Then the allowable bond strength for the adhesive based on a safety factor of four is tabulated along with the allowable load capacities for various threaded steel anchor rods or reinforcing bars. The designer may wish to use higher safety factors than the minimum safety factor of four. ... As in all applications, the actual safety factors used and allowable load capacities should be reviewed and verified by a design professional familiar with the actual product installation."

*Provide final
Report when
issued.*

It is agreed that the factor of safety only applies to the adhesive. The lowest factor of safety that Powers Fasteners would ever recommend, however, is four. The allowable bond strengths in tension and/or shear are then compared to the respective allowable steel strengths, using the lesser of the two values as the allowable load capacity of the anchor. Ultimately, it would have to be the decision of the design professional to reduce the factor of safety down to two. What excerpt from the manufacturer's catalog does the hand written comment refer to?

II) $T_{allow-adh}$ should be equal to 6,350 lbs., $V_{allow-adh}$ should be equal to 4,415 lbs. It is agreed that $T_{allow-rod}$ should be corrected to equal 7,560 lbs. $V_{allow-rod}$ should be equal to 3,915 lbs.

Section C: "Interaction Factors", Four bolt connection loading condition (F.S. = 4.0)

Powers Fasteners' comment: Agree - $T_{allow-adh} = 6,350$ lbs.

Document Title (C09B2) - Adhesive Anchors for Ceiling Supports (Ref. Submittal 723.480-008D)
Page Number 5 - Marked Copy of First Page of Conam Inspection, Inc.'s Heat Testing Report dated November 11, 1993

1) Did test match required embedment depth of 5" minimum?

2) Were any displacements observed while the anchor was under load and with fire applied?

Powers Fasteners' comments: A copy of the entire test report is attached. This test was performed for a similar ceiling panel project. No embedment depths are reflected in the report. However, a proof load of 6,250 lbs. was targeted. The concrete test blocks had a compressive strength of 4,752 psi. Based on analysis of Powers' published performance data current at that time (copy attached), the minimum embedment depth that would have been needed to achieve an allowable tensile load of 6,250 lbs. would have been between 4-7/8" and 5".

Ref. Specification 723.480, page 7 of 13, Section 2.08, "Adhesive Type Ceiling Anchorage System", Paragraph B

Powers Fasteners' comments: Please find attached a Product Profile describing a similar application where Power-Fast Epoxy Injection Gel was used successfully.

Should you have any questions or require further information, please feel free to contact me at (914) 235-6300.

Very truly yours,

Powers Fasteners, Inc.


Ray Williamson
Engineering Manager

cc: William H. Alderman, P.E. - Powers Fasteners, Inc.
Jack Armour - Powers Fasteners, New England

Powers Fasteners, Inc.

2 Powers Square
New Rochelle, New York 10801

Stephen A. Rice
Field Engineering Manager

914-235-6300
Fax: 914-2358774
E-mail: rice@powers.com

Wednesday, October 20, 1999

Mr. Neil Joyce
Modern Continental Construction
6 Neco Court
Boston MA 02210

Re: Powers Data Interpolation (Contract CO9B2 I-93 Tunnel Finishes)

Dear Mr. Joyce:

In response to your fax memo to Mr. Gary Wilcox dated 10/19/99 regarding interpolation and per my discussion with Jack Armour of Powers Fasteners Boston, I have the following response.

Please reference the attached page no.68 from the Second Edition Powers Rawl Fastening System Design Manual. In the last paragraph of section no. 22.6.3 Depth of Embedment, it states, "For applications requiring installation at embedment depths between those published, linear interpolation is permitted."

For further information, please contact me at (914) 235-6300. Thank you for your attention.

Stephen A. Rice
Powers Fasteners

Attachment (p.68 Fastening System Design Manual 2ed.)

Cc: Westwood Systems, Inc., Gary Wilcox.
Powers Fasteners, Inc., William Alderman.
Powers Fasteners Boston, Jack Armour.

*Powers Fastening, Inc. (USA)
New Rochelle, New York
Brewster, New York
Powers Industries Ltd. (Canada)
Mississauga, Ontario
Rawl Australasia, Pty.
Victoria, Australia
Rawl New Zealand, Ltd.
Auckland, New Zealand
Powers Metal Manufacturing Company
Hudson, Michigan
International Pin
Park Forest, Illinois*

Anchor Selection Guidelines

22.6.2 Combined Loading

The combination of loads for anchors loaded in both tension and shear should be proportional as follows based upon the Uniform Building Code:

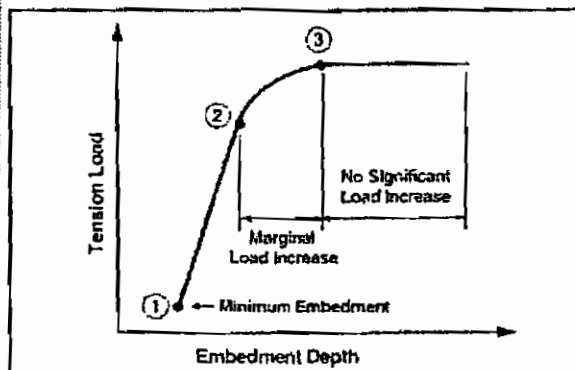
$$\left(\frac{T_s}{T_A}\right)^{5/3} + \left(\frac{V_s}{V_A}\right)^{5/3} \leq 1$$

Where: T_s = Applied Service Tension Load
 T_A = Allowable Tension Load
 V_s = Applied Service Shear Load
 V_A = Allowable Shear Load

22.6.3 Depth Of Embedment

The depth of embedment published for each anchor in the load capacity charts is critical to achieving the expected load capacities. This depth is measured from the surface of the base material to the bottom of the anchor. For mechanical expansion anchors, this would be the depth measured to the bottom of the anchor prior to actuation. For each anchor type, a minimum embedment depth is specified. This depth is typically the minimum required for proper anchor installation and reliable functioning. Attempting to install an anchor at less than the minimum required may overstress the base material causing it to fail when the anchor is expanded. In some masonry materials, the minimum depth may be decreased depending upon the anchor style as noted in the load tables.

As noted in Section 2.8, the load capacity of some anchor types will increase with deeper embedments. For anchors which exhibit this behavior, multiple embedment depths and the corresponding load capacity are listed. As the embedment depth is increased, the load capacity will increase up to a transition point. This point is usually the maximum embedment depth listed. At this point, mechanical anchors may experience material failure or localized failure of the base material around the expansion mechanism. Adhesive type anchors may reach the capacity of the bond, the anchor rod material, or the capacity of the base material. The following diagram shows the typical performance of a mechanical anchor installed in concrete.



At the minimum embedment depth, the mode of failure at the ultimate load capacity is typically a concrete shear cone. As the anchor is installed at a deeper embedment depth, the size of the theoretical concrete shear cone increases, resulting in an increased load capacity. As the embedment depth is increased towards point 2, the mode of failure changes from a concrete shear cone to localized failure around the expansion mechanism. Beyond this point, marginal load capacity increases can be expected until the capacity of the expansion mechanism or anchor material is reached at embedment depths corresponding to point 3. The load capacity will not increase significantly for anchors installed at embedment depths beyond this point. This point is usually the deepest embedment listed in the anchor load capacity tables and is the maximum recommended. Applications which require an embedment deeper than those published should be tested to verify proper anchor performance.

For applications requiring installation at embedment depths between those published, linear interpolation is permitted.

22.6.4 Base Material Strength

As discussed in Section 2.2, the strength of the base materials in which anchors may be installed varies widely and is a key factor in the performance of an anchor. Powers Rawl publishes the average ultimate load capacities for anchors installed in concrete and masonry units along with other appropriate base materials depending upon the product.

For installations in concrete, the load capacity of an anchor usually increases as the compressive strength increases. The load capacities for anchors installed in concrete are published for compressive strengths of 2000, 4000, and 6000 psi. Linear interpolation of the data for intermediate compressive strengths is permissible.

For masonry unit base materials, the published load capacities should be used as a guide since the consistency of these materials varies widely. Job site tests are recommended for critical applications in these materials.

22.6.5 Base Material Thickness

The minimum recommended thickness of solid concrete or masonry base material, BMT, when using a mechanical or adhesive anchor is 125% of the embedment to be used. For example, when installing an anchor to a depth of 4", the base material should be at least 5" thick. Conversely, the maximum embedment should be 80% of the base material thickness. If a concrete slab is 10" thick, an 8" depth would be the maximum recommended anchor embedment. This does not apply to products designed for installation in hollow base materials as noted in the individual anchor sections.

Section 2

Structural Calculations

Structural Calculations

Project: Central Artery

Seismic Code: Mass Bldg Code
(780 CMR)

File: tunnelceilingrev4 ✓

Customer: Modern Continental
Construction Co.

Seismic Zone: II

Date: Dec. 17, 1999

Equipment: Tunnel finishes

Calc's by: JV

Chkd by: JT

A. Design Service Loads (Per project spec. section 723.480 2.08E)

Page 1 of 2

Per section 2.08-E-1. Two bolt connection loading condition

a. Vertical tension force on adhesive = 4000 lbs (F.S = 4)

Vertical tension force on steel rod = 2600 lbs

b. Vertical tension force on adhesive = 6000 lbs (F.S = 2)

Vertical tension force on steel rod = 3900 lbs

Per section 2.08-E-2. Four bolt connection loading condition

a. Vertical tension force on adhesive = 3310 lbs (F.S = 4)

Vertical tension force on steel rod = 2150 lbs

Horizontal shear force applied to adhesive and steel rod = 1000 lbs

B. Allowable Loads (Per Power Rawl 2nd edition section 52 pages 143, 144, 160 & 161)

The allowables are based on the following min. requirements:

Steel rod to be 316 SS

Drill diameter = 3/4"

Concrete strength = 4000 psi

Minimum anchor embedded length = 5"

Minimum anchor spacing (Tension or Shear Load) = $8d = 8 \times 0.625" = 5"$

Minimum edge distance (Tension Load) = $6d = 6 \times 0.625" = 3.75"$

Minimum edge distance (Shear Load) = $12d = 12 \times 0.625" = 7.5"$ Governs



Adhesive allowables (Power-Fast Epoxy Injection Gel)

316 Stainless steel rod allowables

a. Adhesive Tension Allowable = 6350 lbs

c. Steel Rod Tension Allowable = 7560 lbs

b. Adhesive Shear Allowable = 4415 lbs

d. Steel Rod Shear Allowable = 3915 lbs

C. Interaction factors $((T_{actual} / T_{allowable}) + (V_{actual} / V_{allowable})) < 1.0$



1. Two bolt connection loading condition (loads per section 2.08-E-1a)

$T_{adh} = 4000$

$V_{adh} = 0$

$T_{allow_adh} = 6350$

$V_{allow_adh} = 4415$

$T_{rod} = 2600$

$V_{rod} = 0$

$T_{allow_rod} = 7560$

$V_{allow_rod} = 3915$

$IF_{adh} = \frac{T_{adh}}{T_{allow_adh}} + \frac{V_{adh}}{V_{allow_adh}}$

$IF_{adh} = 0.63$ ✓ < 1.0 O.K

$IF_{rod} = \frac{T_{rod}}{T_{allow_rod}} + \frac{V_{rod}}{V_{allow_rod}}$

$IF_{rod} = 0.344$ < 1.0 O.K

Structural Calculations

Page 2 of 2

4

2. Two bolt connection loading condition (loads per section 2.08-E-1b)

Tadh = 6000 Vadh = 0 Tallow_adh = 6350 Vallow_Adh = 4415
Trod = 3900 Vrod = 0 Tallow_rod = 7560 Vallow_rod = 3915

$$IFadh = \frac{Tadh}{Tallow_adh} + \frac{Vadh}{Vallow_Adh}$$

$$IFadh = 0.945 \quad \checkmark < 1.0 \text{ O.K.} \quad \checkmark$$

$$IFrod = \frac{Trod}{Tallow_rod} + \frac{Vrod}{Vallow_rod}$$

$$IFrod = 0.516 < 1.0 \text{ O.K.}$$

4

3. Four bolt connection loading condition (loads per section 2.08-E-2a)

Tadh = 3310 Vadh = 1000 Tallow_adh = 6350 Vallow_adh = 4415
Trod = 2150 Vrod = 1000 Tallow_rod = 7560 Vallow_rod = 3915

$$IFadh = \left(\frac{Tadh}{Tallow_adh} \right) + \left(\frac{Vadh}{Vallow_adh} \right)$$

$$IFadh = 0.748 \quad \checkmark < 1.0 \text{ O.K.}$$

$$IFrod = \left(\frac{Trod}{Tallow_rod} \right) + \left(\frac{Vrod}{Vallow_rod} \right)$$

$$IFrod = 0.54 < 1.0 \text{ O.K.}$$

NOTE

1. These calculations were performed to compare the anchor minimum design service loads, per project specifications section 723.480 2.08E, with the anchor allowable loads, provided by the anchor bolt manufacturer only.



(SEE NOTE.)

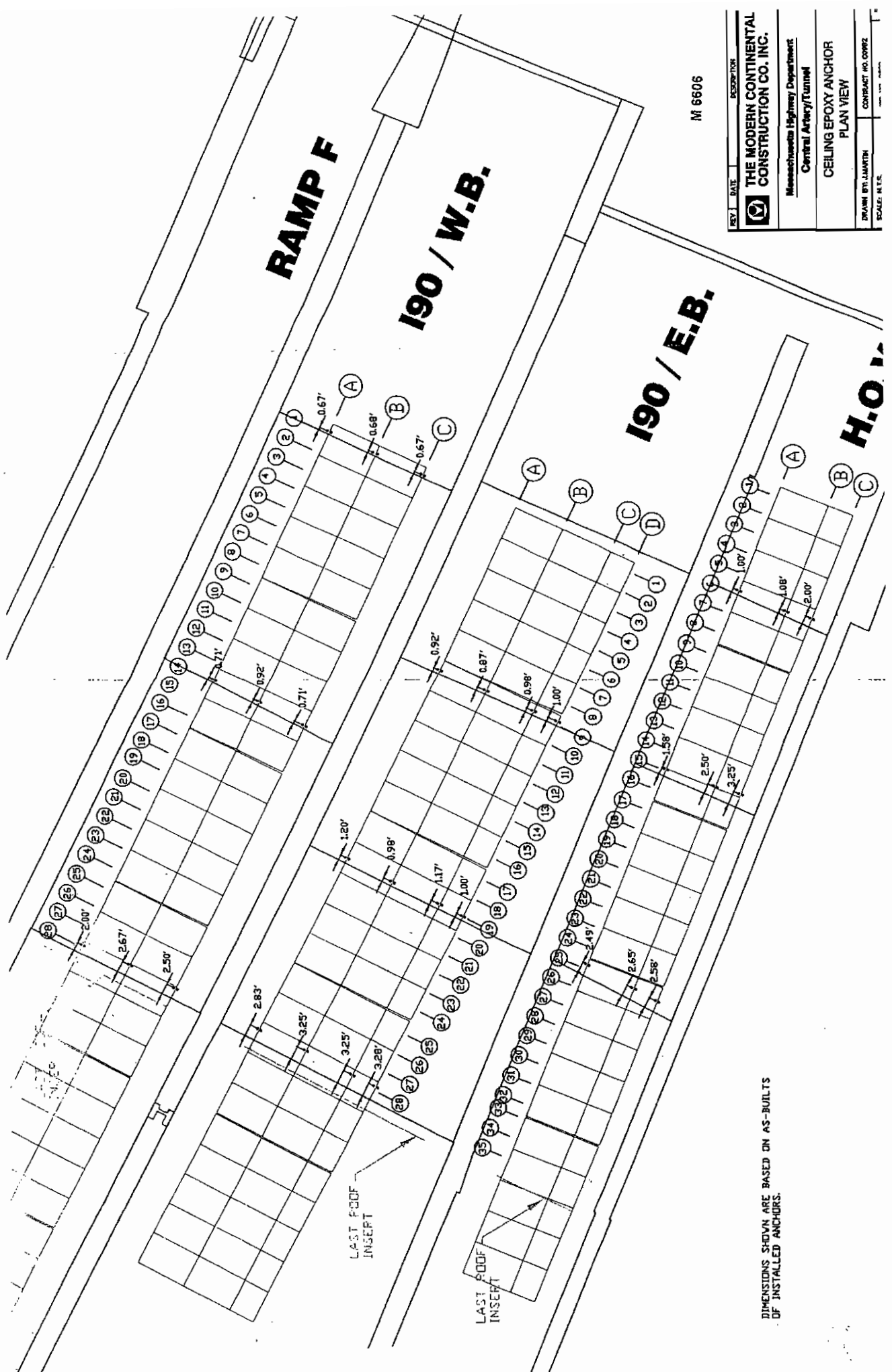
Section 3

Edge Distance Confirmation

Contract C09B2 -I-90 Seaport Access Tunnel Finishes

Reference Submittal No. 723.480 – 014 C
 Structural Calculations for Adhesive Anchors

Per the Manufacturer's recommended minimum edge distance, and in accordance with Contract Drawing S-387, Section A, the attached drawing clarifies the as-built condition of the adhesive anchors installed in the existing tunnel constructed under Contract C04A2. The minimum edge distance (distance from construction joint to center line of adjacent anchor bolts) is confirmed at eight (8) inches (0.67 feet).



DIMENSIONS SHOWN ARE BASED ON AS-BUILTS
OF INSTALLED ANCHORS.

M 6606

REV	DATE		THE MODERN CENTINETAL CONSTRUCTION CO. INC.	PROJECT NO.
Massachusetts Highway Department Central Artery/Tunnel			CONTRACT NO. 00962	DRAWN BY: JAVITTIN SCALE: N.T.S.
CEILING EPOXY ANCHOR PLAN VIEW				

Section 4

Draft Copy of Engineering Report

**Power's Concrete Anchors
Report No ER-4514
October 1, 1999**

SUBJECT: POWERS CONCRETE ANCHORS**REPORT NO.** ER-4514**COMPANY:** POWERS FASTENERS, INC.**DATE:****ADDRESS:** TWO POWERS SQUARE
NEW ROCHELLE, NEW YORK 10801

Reissued October 1, 1999

DRAFT**1.0 SUBJECT:**

Power-Fast Epoxy and Chem-Stud Capsule Adhesive Anchor Systems

2.0 DESCRIPTION:**2.1 Power-Fast:****2.1.1 General:**

The Power-Fast Adhesive Anchor System consists of a two-component structural epoxy that is used to bond threaded rods, fully threaded bolts, or reinforcing steel into predrilled holes in normal-weight, stone aggregate concrete.

2.1.2 Materials:

2.1.2.1 Adhesive: Power-Fast adhesive is a two-component structural epoxy that is packaged in equal volume side-by-side plastic cartridges or pumps. The components include a base resin and hardener that are mixed in a one-to one ratio as they are dispensed through a disposable static mixing nozzle or manually in a pump cartridge. Fast Set and Standard Set versions are available. The minimum shelf life of the cartridges is two years when stored at temperatures between 40° F to 90° F (4.4° C to 32.2° C).

2.1.2.2 Anchor Rods: Threaded rods range in diameter from 3/8 to 1-1/4 inches (9.5 to 31.8mm) and are manufactured from carbon steel in compliance with the mechanical properties of ASTM A 36, ASTM A 307, or ASTM A 193 Grade B7.

Stainless steel threaded rod is manufactured from AISI Types 304 or 316 material.

Fully threaded bolts (Chem-Bolt) range in diameter from 3/8 to 3/4 inches (9.5 to 19.1mm) and are manufactured from carbon steel in compliance with the mechanical properties of SAE Grade 2 or Grade 5. Fully threaded stainless steel bolts are manufactured from AISI Type 304 material. Specification and installation requirements for threaded rod are listed in Table 1.

2.1.2.3 Reinforcing Bars: Deformed reinforcing bars range in size from No. 3 to No. 10. The bars are manufactured from steel conforming to ASTM A 615, A 616, A 617, or A 706. Specification and installation requirements for reinforcing bars are listed in Table 8.

2.1.2.4 Concrete: Anchors are installed in normal-weight, stone aggregate concrete having various compressive strengths as listed in Tables 4, 5, 6, 7, 10, and 11.

2.2 Chem-Stud

2.2.1 General:

The Chem-Stud Anchor System consists of a two-part resin capsule that is used to bond chisel pointed threaded rods into predrilled holes in normal-weight, stone aggregate concrete.

2.2.2 Materials:

2.2.2.1 Adhesive: The Chem-Stud capsule contains pre-measured amounts of resin and hardener in two sealed glass tubes. The outer glass tube is filled with vinyl ester resin and quartz aggregate. A smaller sealed glass tube is suspended in the resin and is filled with a hardening catalyst. The adhesive components are

mixed by spinning a chisel pointed anchor rod into the capsule using a rotary hammer. A specific capsule size is provided for each diameter of anchor rod. The minimum shelf life of the capsules is two to four years when stored at temperatures under 71° F (21.1° C).

2.2.2.2 Anchor Rods: Chem-Stud threaded anchor rods have a chisel point for proper mixing of the adhesive. Threaded rods range in diameter from 3/8 to 1-1/4 inches (9.5 to 31.8mm) and are manufactured from carbon steel in compliance with the mechanical properties of ASTM A 36, ASTM A 307, or ASTM A 193 Grade B7. Stainless steel threaded rod is manufactured from AISI Types 304 or 316 material. Specification and installation requirements for threaded rod are listed in Table 12.

2.2.2.3 Concrete: Anchors are installed in normal-weight, stone aggregate concrete with various compressive strengths as listed in Tables 15 and 16.

2.3 Design

Allowable static loads for anchors installed in accordance with this report are listed in Tables 4, 5, 6, 7, 10, 11, 15, and 16. The allowable tension loads for the adhesives must be adjusted for elevated temperatures in accordance with Figures 1 and 2. Allowable loads for anchors subjected to combined shear and tension forces are determined by the following equation:

$$(P_s/P_t) + (V_s/V_t) \leq 1$$

where:

P_s = Applied service tension load

P_t = Service tension load

V_s = Applied service shear load

V_t = Service shear load

Use of these anchors in resisting vibratory or shock loads, such as those present in supports for reciprocating engines or crane loads, is beyond the scope of this report.

2.4 Installation:

2.4.1 General: Anchors shall be installed according to the manufacturer's published installation instructions and the requirements of this report. Anchor holes must be pre-drilled in normal weight concrete with a bit meeting the dimensions listed in ANSI Standard B 212.15-1994. Anchor holes should be drilled to the specified size and embedment depth and cleaned using compressed air and a nylon brush. Hole diameter, embedment, spacing, edge distance, and concrete requirements are shown in Tables 3, 9, and 14.

2.4.2 Power-Fast: The Power-Fast epoxy is injected into the hole, starting at the bottom until the hole is approximately one-half full. The threaded rod, fully threaded bolt, or reinforcing bar is then inserted, while rotating, until fully embedded in the anchor hole. The Power-Fast epoxy must be allowed to cure in accordance with Table 2 prior to the application of the design load.

2.4.3 Chem-Stud: The adhesive capsule is inserted completely into the anchor hole. Using a rotary hammer, the chisel-pointed threaded rod is spun into the adhesive capsule until fully embedded in the anchor hole. The Chem-Stud adhesive must be allowed to cure in accordance with Table 13 prior to the application of the design load.

DRAFT

2.4.4 Special Considerations: The anchors may be used within fire-resistive construction, provided the anchors only resist wind and/or seismic forces. The anchors cannot be used to resist tension loads in ceiling or wall installations unless special consideration is given to fire-exposure conditions.

2.5 Special Inspection:

Adhesive anchor installations require special inspection in accordance with Section 1701 of the code. The special inspector must record the product description including adhesive expiration date, concrete type and strength, drill bit dimensions in compliance with this report, hole depth and cleanliness, anchor rod diameter and length, steel grade, and verify that the anchor installation is in accordance with the manufacturer's published installation instructions.

2.6 Identification:

The anchors are identified in the field by the packaging that indicates the manufacturer's name (Powers Fasteners), product name, material type, batch number, and evaluation report number (ICBO ES ER-4514).

3.0 EVIDENCE SUBMITTED:

Data complying with the ICBO ES Acceptance Criteria for Adhesive Anchors in Concrete and Masonry Elements (AC508), dated January 1999.

4.0 FINDINGS:

That the Powers concrete anchors described in this report comply with the 1997 *Uniform Building Code*TM, subject to the following conditions:

- 4.1 The Power Fast Epoxy System, Standard Set is permitted to be used for long-term live load, dead load, and short-term loads such as those resulting from wind or earthquake forces. The Power Fast Epoxy System, Fast Set is permitted to be used for short-term loads such as those resulting from wind or earthquake forces.
- 4.2 The allowable loads may be increased by 33-1/3 percent for short term loading due to seismic or wind forces as noted in Tables 4, 5, 6, 7, 10, 11, 15, and 16.
- 4.3 The anchors are installed in accordance with the manufacturer's instructions and this report.
- 4.4 Anchors are installed in holes pre-drilled using a carbide tipped drill manufactured within the range of maximum and minimum drill tip dimensions according to ANSI B212.15-1994.
- 4.5 Calculations justifying that the applied loads comply with this report are submitted to the building official for approval.
- 4.6 Special inspection is provided as set forth in Section 2.5 of this report.
- 4.7 Due to the lack of an accepted test method and procedure for the evaluation of anchors in cracked concrete, anchors are limited to installation in uncracked concrete, which is concrete subjected to tensile stresses not exceeding 170 psi (1.2 MPa) as induced by external loads or deformations.
- 4.8 Due to the lack of an accepted test method and procedure for evaluating data to determine the performance of adhesive of anchors subjected to fatigue and/or shock loading, use of these anchors to resist vibratory or shock loads, such

as those present in supports for reciprocating engines or crane loads, is beyond the scope of this report.

4.9 Use of these anchors with unprotected carbon steel is limited to dry, interior exposure. Installations exposed to severe, moderate, or negligible weathering conditions, as defined by Figure 21-1-1 of UBC Standard 21-1, are permitted where stainless steel anchor rods are used.

4.10 Anchors are manufactured by the Powers Fasteners, Inc., Two Powers Square, New Rochelle, N.Y. with quality control inspections by CEL Consulting (AA-639).

This report is subject to re-examination in one year.

TABLE 1 - SPECIFICATIONS AND DETAILS FOR INSTALLATION OF THREADED ROD IN CONCRETE USING POWER-FAST EPOXY ADHESIVE

PROPERTY	ROD DIAMETER (Inches)						
	3/8	1/2	5/8	3/4	7/8	1	1-1/4
A _n (Nominal area of rod (in. ²))	0.1105	0.1963	0.3068	0.4418	0.6013	0.7854	1.2772
BD (Nominal bit diameter (in.))	7/16	9/16	3/4	7/8	1	1 1/8	1 3/8
E (Minimum anchor embedment (in.))	1 1/2	2	2 1/2	3	3 1/2	4	5
T (Maximum tightening torque (ft.-lb.))	20	40	90	160	280	380	600

For SI: 1 inch = 25.4 mm, 1 square inch = 645.16 mm², 1 ft.-lb. = 1.35 N-m

TABLE 2 - MANUFACTURER'S RECOMMENDED CURE TIME FOR POWER-FAST EPOXY ADHESIVE

BASE MATERIAL TEMPERATURE ¹ (°F)	MAXIMUM GEL TIME ² (minutes)		MINIMUM CURING TIME ³ (hours)		FULL CURING TIME ⁴ (hours)	
	Fast Set	Std. Set	Fast Set	Std. Set	Fast Set	Std. Set
40	30	60	2	16	36	48
60	20	45	3	7	24	36
75	15	35	1	6	24	24
90	10	20	1-1/2	4	16	24

For SI: °C = 5/9(°F - 32)

¹This is the temperature as measured in the base material.

²The gel time is the maximum time during which the epoxy can be dispensed before it begins to set.

³Anchors should not be disturbed during the minimum curing time. When the minimum cure time is achieved, the anchors can sustain a design load and the fixture can be positioned.

⁴The full curing time is the minimum time required for the epoxy to achieve its ultimate load capacities.

TABLE 3 - REDUCTIONS FOR SPACING AND EDGE DISTANCES FOR THREADED ROD OR FULLY THREADED BOLTS INSTALLED IN CONCRETE WITH POWER-FAST EPOXY ADHESIVE

DIMENSION / LOAD TYPE	DISTANCE FOR FULL ANCHOR CAPACITY ¹	MINIMUM DISTANCE ²	REDUCTION FACTOR ³
SPACING (s) BETWEEN ANCHORS - TENSION AND SHEAR LOADS	s _{cr} = 16D	s _{min} = 8D	F _t = F _v = 0.72
EDGE DISTANCE (e) - TENSION LOADS	e _{cr} = 18D	e _{min} = 4D	F _t = 0.56
EDGE DISTANCE (e) - SHEAR LOADS	e _{cr} = 12D	e _{min} = 4D	F _v = 0.21

¹The listed values are the minimum distances required to obtain the load values listed in Table Nos. 4, 5, 6, and 7. D = Anchor Diameter. When adjacent anchors are different sizes or embedments, use largest value for D.

²The listed values are the minimum distances at which the anchor can be set, when load values are adjusted appropriately.

³Load values in the tables are multiplied by the reduction factor when anchors are installed at the minimum spacing listed. Use linear interpolation for spacing between critical and minimum distances. Multiple reduction factors for more than one spacing or edge distance are calculated separately and multiplied.

TABLE 4 - ALLOWABLE TENSION LOADS FOR THREADED RODS INSTALLED IN NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE 1, 2, 3, 4, 5, 6

ANCHOR DIAMETER (inches)	BIT DIAMETER (inches)	SPACING, s (inches)	EDGE, c (inches)	MINIMUM EMBED. (inches)	ALLOWABLE TENSION LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds)				ALLOWABLE TENSION LOAD BASED ON STEEL STRENGTH (pounds)			
					$f_c=2000$ psi	$f_c=3000$ psi	$f_c=4000$ psi	$f_c=5000$ psi	ASTM A 36	ASTM A 307	ASTM A 193 B7	304/316 SS
3/8	7/16	6	3 3/4	1 1/2	863	980	1,110	1,300	2,115	2,185	4,555	3,630
				1 7/8	1,170	1,336	1,477	1,666				
				2 1/4	1,475	1,692	1,844	2,032				
				2 5/8	1,780	2,048	2,211	2,398				
				3	2,085	2,404	2,578	2,764				
				3 3/8	2,390	2,760	2,945	3,130				
				3 3/4	2,612	2,986	3,125	3,286				
				4 1/8	2,834	3,212	3,305	3,442				
				4 1/2	3,056	3,438	3,483	3,598				
				4 7/8	3,278	3,664	3,665	3,754				
1/2	9/16	8	3	3 1/4	3,500	3,890	3,845	3,910	3,755	3,883	8,095	6,470
				3	1,200	1,810	2,705	3,615				
				2 1/2	1,609	2,128	3,107	3,980				
				3	2,018	2,446	3,509	4,345				
				3 1/2	2,427	2,764	3,911	4,710				
				4	2,836	3,082	4,313	5,075				
				4 1/2	3,245	3,400	4,715	5,440				
				5	3,640	3,939	5,094	5,777				
				5 1/2	4,035	4,478	5,473	6,114				
				6	4,430	5,017	5,832	6,451				
5/8	3/4	10	6 1/4	6 1/2	4,825	5,356	6,231	6,788	5,870	6,075	12,655	10,130
				7	5,220	6,093	6,610	7,125				
				2 1/2	1,725	2,070	2,350	2,625				
				3 1/8	2,347	2,768	3,050	3,329				
				3 3/4	2,969	3,466	3,730	4,033				
				4 3/8	3,591	4,164	4,450	4,737				
				5	4,213	4,862	5,150	5,441				
				5 5/8	4,835	5,560	5,850	6,145				
				6 1/4	5,409	6,050	6,318	6,590				
				6 7/8	5,983	6,540	7,186	7,515				
3/4	7/8	12	7 1/2	7 1/2	6,557	7,030	7,854	8,200	8,455	8,750	18,225	12,400
				8 1/8	7,131	7,520	8,522	8,885				
				8 3/4	7,705	8,010	9,190	9,570				
				3	2,400	3,030	3,710	4,360				
				3 3/4	3,328	4,114	4,844	5,642				
				4 1/2	4,256	5,178	5,978	6,924				
				5 1/4	5,184	6,242	7,112	8,206				
				6	6,112	7,308	8,246	9,488				
				6 3/4	7,040	8,370	9,380	10,770				
				7 1/2	7,960	9,472	10,488	11,106				
7/8	1	14	8 3/4	8 1/4	8,880	9,574	10,396	11,442	11,510	11,905	24,805	16,850
				9	9,780	10,176	10,904	11,778				
				9 3/4	10,680	10,778	11,412	12,114				
				10 1/2	11,580	11,390	11,920	12,450				
				3 1/2	2,730	3,330	3,895	4,455				
				4 3/8	3,818	4,564	5,224	5,859				
				5 1/4	4,905	5,838	6,553	7,263				
				6 1/8	5,994	7,092	7,882	8,667				
				7	7,082	8,346	9,211	10,071				
				7 7/8	8,170	9,600	10,540	11,475				

TABLE 4 - ALLOWABLE TENSION LOADS FOR THREADED RODS INSTALLED IN NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE (Continued) 1, 2, 3, 4, 5, 6

ANCHOR DIAMETER (inches)	BIT DIAMETER (inches)	SPACING, s (inches)	EDGE, c (inches)	MINIMUM EMBED. (inches)	ALLOWABLE TENSION LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds)				ALLOWABLE TENSION LOAD BASED ON STEEL STRENGTH (pounds)			
					$P_c=1000 \text{ psi}$	$P_c=3000 \text{ psi}$	$P_c=4000 \text{ psi}$	$P_c=5000 \text{ psi}$	ASTM A 36	ASTM A 307	ASTM A 193 B7	304/316 SS
1	1-1/8	16	10	4	3,385	4,085	4,695	5,300	15,025	13,550	22,400	22,020
				5	4,696	5,694	6,593	7,486				
				6	6,067	7,303	8,491	9,672				
				7	7,518	8,912	10,389	11,858				
				8	8,629	10,521	12,287	14,044				
				9	9,940	12,130	14,185	16,230				
				10	10,897	13,142	15,078	17,005				
				11	11,834	14,154	15,971	17,780				
				12	12,811	15,166	16,864	18,553				
				13	13,768	16,178	17,757	19,330				
				14	14,725	17,190	18,650	20,105				
				5	4,500	6,020	7,690	9,695	23,485	24,295	50,615	34,420
1-1/4	1-3/8	20	12 1/2	6 1/4	6,246	8,156	10,348	12,901				
				7 1/2	7,992	10,392	13,006	15,907				
				8 3/4	9,738	12,428	15,664	19,013				
				10	11,484	14,364	18,322	22,119				
				11 1/4	13,230	16,700	20,980	25,233				
				12 1/2	15,138	18,822	22,948	27,842				
				13 3/4	17,086	20,944	24,916	28,859				
				15	19,014	23,066	26,884	30,676				
				16 1/4	20,942	25,188	28,852	32,493				
				17 1/2	22,870	27,310	30,820	34,310				

For SI: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength. When using the Fast Set version, the bond strength must be reduced by 25%. The steel strength values listed are in accordance with Section 2.1.2.2.

²Allowable loads may be increased by 33-1/3% for short-term loading due to earthquakes or wind.

³The tabulated values are for anchors installed with edge distance (c) and spacing (s) in accordance with Table 3. The reduction factors listed in this table shall be applied to the values listed in the bond or concrete strength column.

⁴The tabulated shear and tension values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation. Linear interpolation may be used for concrete strengths between those listed.

⁵Allowable loads based on bond strength are calculated based on a safety factor of 4.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

TABLE 5 - ALLOWABLE SHEAR LOADS FOR THREADED RODS INSTALLED IN NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE 1, 2, 3, 4, 5, 6

ANCHOR DIAMETER (inches)	BIT DIAMETER (inches)	SPACING, s (inches)	EDGE, c (inches)	MINIMUM EMBED. (inches)	ALLOWABLE SHEAR LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds)	ALLOWABLE SHEAR LOAD BASED ON STEEL STRENGTH (pounds)			
						ASTM A 36	ASTM A 307	ASTM A 193 B7	304/316 SS
3/8	7/16	6	4 1/2	1 1/2	1,150	1,090	1,125	2,345	1,870
				1 7/8	1,320				
				2 1/4	1,250				
				2 5/8	1,360				
				3	1,430				
				3 3/8	1,500				
1/2	9/16	8	6	2	1,700	1,940	2,000	4,170	3,330
				2 1/2	2,000				
				3	2,380				
				3 1/2	2,600				
				4	2,900				
				4 1/2	3,200				
5/8	3/4	10	7 1/2	2 1/2	2,400	3,025	3,130	6,520	5,210
				2 7/8	3,060				
				3 3/4	3,720				
				4 3/8	4,380				
				5	5,040				
				5 5/8	5,700				

TABLE 5 -- ALLOWABLE SHEAR LOADS FOR THREADED RODS INSTALLED IN NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE (Continued) 1, 2, 3, 4, 5, 6

ANCHOR DIAMETER (Inches)	BIT DIAMETER (Inches)	SPACING, s (Inches)	EDGE, e (Inches)	MINIMUM EMBED. (Inches)	ALLOWABLE SHEAR LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds) $F_c \geq 2000$ psi Minimum	ALLOWABLE SHEAR LOAD BASED ON STEEL STRENGTH (pounds)			
						ASTM A 36	ASTM A 307	ASTM A 193 B7	304/316 89
3/4	7/8	12	9	3	3,600	4,355	4,507	9,390	6,390
				3 3/4	4,160				
				4 1/2	4,720				
				5 1/4	5,280				
				6	5,840				
				6 3/4	6,400				
7/8	1	14	10 1/2	3 1/2	3,900	5,930	6,133	12,780	8,680
				4 3/8	4,640				
				5 1/4	5,780				
				6 1/8	6,920				
				7	8,060				
				7 7/8	9,200				
1	1 1/8	16	12	4	4,600	7,745	8,010	16,690	11,340
				5	6,180				
				6	7,760				
				7	9,340				
				8	10,920				
				9	12,500				
1 1/4	1 3/8	20	15	5	5,300	12,100	12,515	26,075	17,720
				6 1/4	8,090				
				7 1/2	10,500				
				8 3/4	13,000				
				10	15,500				
				11 1/4	18,000				

For SI: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength. The steel strength values listed are in accordance with Section 2.1.2.2.

²For short-term loading due to earthquakes or wind allowable shear loads for A 36 & A 307 may be increased by 33-1/3%. A 36 & A 307 values must be used instead of listed values for higher strength steels for these loading conditions.

³The tabulated values are for anchors installed with edge distance (e) and spacing (s) in accordance with Table 3. The reduction factors listed in this table shall be applied to the values listed in the bond or concrete strength column.

⁴The tabulated load values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation. Linear interpolation may be used for concrete strengths between those listed.

⁵Allowable loads based on bond strength are calculated based on a safety factor of 4.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

TABLE 6 - ALLOWABLE TENSION LOADS FOR FULLY THREADED BOLTS INSTALLED IN
NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE 1, 2, 3, 4, 5, 6, 7

ANCHOR DIAMETER (inches)	BIT DIAMETER (inches)	SPACING, s (inches)	EDGE, c (inches)	MINIMUM EMBED. (inches)	ALLOWABLE TENSION LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds)				ALLOWABLE TENSION LOAD BASED ON STEEL STRENGTH (pounds)			
					$f_c=2000$ psi	$f_c=3000$ psi	$f_c=4000$ psi	$f_c=5000$ psi	GRADE 1	GRADE 2	GRADE 3	304/316 SS
3/8	7/16	8	3 3/4	1 1/2	853	988	1,110	1,300	2,180	2,685	4,355	3,630
				1 7/8	1,170	1,336	1,477	1,666				
				2 1/4	1,473	1,692	1,844	2,032				
				2 5/8	1,780	2,048	2,211	2,398				
				3	2,083	2,404	2,578	2,764				
				3 3/8	2,396	2,760	2,945	3,130				
				3 3/4	2,612	2,988	3,123	3,266				
				4 1/8	2,834	3,212	3,309	3,442				
				4 1/2	3,056	3,438	3,489	3,598				
				4 7/8	3,278	3,664	3,663	3,754				
				5 1/4	3,500	3,890	3,843	3,910				
				2	1,200	1,810	2,703	3,613	2,880	4,765	7,760	6,470
1/2	9/16	8	5	2 1/2	1,609	2,128	3,107	3,980				
				3	2,018	2,448	3,509	4,343				
				3 1/2	2,427	2,764	3,911	4,710				
				4	2,836	3,082	4,313	5,073				
				4 1/2	3,245	3,400	4,715	5,440				
				5	3,640	3,939	5,094	5,777				
				5 1/2	4,035	4,478	5,473	6,114				
				6	4,430	5,017	5,852	6,451				
				6 1/2	4,823	5,556	6,231	6,788				
				7	5,220	6,095	6,810	7,123				
5/8	3/4	10	6 1/4	2 1/2	1,725	2,070	2,350	2,623	6,080	7,495	12,155	10,130
				3 1/8	2,347	2,768	3,050	3,329				
				3 3/4	2,969	3,466	3,758	4,033				
				4 3/8	3,591	4,164	4,450	4,727				
				5	4,213	4,863	5,150	5,441				
				5 3/8	4,833	5,560	5,850	6,145				
				6 1/4	5,409	6,030	6,318	6,608				
				6 7/8	5,983	6,540	7,186	7,513				
				7 1/2	6,557	7,030	7,854	8,200				
				8 1/8	7,131	7,520	8,523	8,883				
				8 3/4	7,703	8,010	9,190	9,570				
3/4	7/8	12	7 1/2	3	2,400	3,050	3,710	4,360	8,790	10,795	17,500	12,408
				3 3/4	3,328	4,114	4,844	5,642				
				4 1/2	4,236	5,178	5,978	6,924				
				5 1/4	5,184	6,242	7,112	8,206				
				6	6,112	7,306	8,246	9,488				
				6 3/4	7,040	8,370	9,380	10,770				
				7 1/2	7,620	9,072	9,888	11,106				
				8 1/4	8,200	9,874	10,396	11,442				
				9	8,788	10,176	10,904	11,778				
				9 3/4	9,368	10,778	11,412	12,114				

For SI: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength. When using the Fast Set version, the bond strength must be reduced by 25%. The steel strength values listed are in accordance with Section 2.1.2.2.

²Allowable loads may be increased by 33-1/3% for short-term loading due to earthquakes or wind.

³The tabulated values are for anchors installed with edge distance (e) and spacing (s) in accordance with Table 3. The reduction factors listed in this table shall be applied to the values listed in the bond or concrete strength column.

⁴The tabulated shear and tension values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation. Linear interpolation may be used for concrete strengths between those listed.

⁵Allowable loads based on bond strength are calculated based on a safety factor of 4.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

⁷According to SAE, Grade 2 requirements for 3/8" to 3/4" diameters apply only to bolts 6" and shorter. For bolts longer than 6", Grade 1 steel requirements apply.

TABLE 7 - ALLOWABLE SHEAR LOADS FOR FULLY THREADED BOLTS INSTALLED IN NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE 1, 2, 3, 4, 5, 6, 7

ANCHOR DIAMETER (inches)	BOLT DIAMETER (inches)	SPACING, s (inches)	EDGE, c (inches)	MINIMUM EMBED. (inches)	ALLOWABLE SHEAR LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds)	ALLOWABLE SHEAR LOAD BASED ON STEEL STRENGTH (pounds)			
					$f'_c = 2000$ psi Minimum	GRADE 1	GRADE 2	GRADE 5	304 / 316 SS
3/8	7/16	6	4 1/2	1 1/2	1,150	1,120	1,385	2,245	1,878
				1 7/8	1,220				
				2 1/4	1,290				
				2 5/8	1,360				
				3	1,430				
				3 3/8	1,500				
1/2	9/16	8	6	2	1,700	1,300	2,465	3,990	3,330
				2 1/2	2,000				
				3	2,300				
				3 1/2	2,600				
				4	2,900				
				4 1/2	3,200				
5/8	3/4	10	7 1/2	2 1/2	2,400	3,130	3,860	4,260	5,210
				3 1/8	3,060				
				3 3/4	3,720				
				4 3/8	4,380				
				5	5,040				
				5 5/8	5,700				
3/4	7/8	12	9	3	3,600	4,518	5,560	9,015	6,390
				3 3/4	4,160				
				4 1/2	4,720				
				5 1/4	5,280				
				6	5,840				
				6 3/4	6,400				

For SI: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength. The steel strength values listed are in accordance with Section 2.1.2.2.

²For short-term loading due to earthquakes or wind allowable shear loads for Grade 1 or Grade 2 may be increased by 33-1/3%. Grade 1 or Grade 2 values must be used instead of listed values for higher strength steels for these loading conditions.

³The tabulated values are for anchors installed with edge distance (c) and spacing (s) in accordance with Table 3. The reduction factors listed in this table shall be applied to the values listed in the bond or concrete strength column.

⁴The tabulated load values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation. Linear interpolation may be used for concrete strengths between those listed.

⁵Allowable loads based on bond strength are calculated based on a safety factor of 4.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

⁷According to SAE, Grade 2 requirements for 3/8" to 3/4" diameters apply only to bolts 6" and shorter. For bolts longer than 6", Grade 1 steel requirements apply.

TABLE 8 - SPECIFICATIONS AND DETAILS FOR INSTALLATION OF REINFORCING BAR IN CONCRETE USING POWER-FAST EPOXY ADHESIVE

PROPERTY	REINFORCING BAR SIZE							
	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	No. 10
A _{sr} [Nominal area of rebar (in. ²)]	0.110	0.200	0.310	0.440	0.600	0.790	1.000	1.270
BD [Nominal bit diameter (in.)]	1/2	3/8	3/4	7/8	1	1 1/8	1 3/8	1 1/2
E [Minimum anchor embedment (in.)]	3 3/8	4 1/2	5 5/8	6 3/4	7 7/8	9	10 1/8	11 1/4

For SI: 1 inch = 25.4 mm, 1 square inch = 645.16 mm²

TABLE 9 - REDUCTIONS FOR SPACING AND EDGE DISTANCES FOR REINFORCING BARS INSTALLED IN CONCRETE WITH POWER-FAST EPOXY ADHESIVE

DIMENSION/LOAD TYPE	DISTANCE FOR FULL ANCHOR CAPACITY ¹	MINIMUM DISTANCE ²	REDUCTION FACTOR ³
SPACING (s) BETWEEN ANCHORS - TENSION AND SHEAR LOADS	s _{cr} = 18D	s _{min} = 9D	F _t = F _v = 0.50
EDGE DISTANCE (c) - TENSION LOADS	c _{cr} = 12D	c _{min} = 6D	F _t = 0.56
EDGE DISTANCE (c) - SHEAR LOADS	c _{cr} = 16D	c _{min} = 4D	F _v = 0.17

¹The listed values are the minimum distances required to obtain the load values listed in Table Nos. 10 and 11. D = Anchor Diameter = Reinforcing Bar in inches. When adjacent anchors are different sizes or embedments, use largest value for D.²The listed values are the minimum distances at which the anchor can be set, when load values are adjusted appropriately.³Load values in the tables are multiplied by the reduction factor when anchors are installed at the minimum spacing listed. Use linear interpolation for spacing between critical and minimum distances. Multiple reduction factors for more than one spacing or edge distance are calculated separately and multiplied.

TABLE 10 - ALLOWABLE TENSION LOADS FOR REINFORCING BARS INSTALLED IN NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE 1, 2, 3, 4, 5, 6

REINFORCING BAR SIZE	BIT DIAMETER (inches)	SPACING, s (inches)	EDGE, c (inches)	MINIMUM EMBED. (inches)	ALLOWABLE TENSION LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds)	ALLOWABLE TENSION LOAD BASED ON STEEL STRENGTH (pounds)	
					f _c = 2000 psi Minimum	GRADE 40	GRADE 60
No. 3	1/2	6 3/4	4 1/2	3 3/8	2,785	2,200	2,640
No. 4	5/8	9	6	4 1/2	3,610	4,000	4,800
No. 5	3/4	11 1/4	7 1/2	5 5/8	5,030	6,200	7,440
No. 6	7/8	13 1/2	9	6 3/4	6,580	8,800	10,560
No. 7	1	15 3/4	10 1/2	7 7/8	8,330	12,000	14,400
No. 8	1 1/8	18	12	9	12,085	15,800	18,960
No. 9	1 3/8	20 1/4	13 1/2	10 1/8	13,930	19,800	23,760
No. 10	1 1/2	22 1/2	15	11 1/4	18,515	24,450	29,450

For SI: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength. When using the Fast Set version, the bond strength must be reduced by 25%. The steel strength values listed are in accordance with Section 2.1.2.3.²Allowable loads may be increased by 33-1/3% for short-term loading due to earthquakes or wind.³The tabulated values are for anchors installed with edge distance (c) and spacing (s) in accordance with Table 9. The reduction factors listed in this table shall be applied to the values listed in the bond or concrete strength column.⁴The tabulated shear and tension values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation. Linear interpolation may be used for concrete strengths between those listed.⁵Allowable loads based on bond strength are calculated based on a safety factor of 4.⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

TABLE 11 – ALLOWABLE SHEAR LOADS FOR REINFORCING BARS INSTALLED IN MINIMUM 2000 PSI NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE 1, 2, 3, 4, 5, 6

REINFORCING BAR SIZE	BIT DIAMETER (inches)	SPACING, s (inches)	EDGE, c (inches)	MINIMUM EMBED. (inches)	ALLOWABLE SHEAR LOAD BASED ON STEEL STRENGTH (pounds)	
					GRADE 40	GRADE 60
No. 3	1/2	6 3/4	6	3 3/8	1,680	1,688
No. 4	5/8	9	8	4 1/2	3,060	3,060
No. 5	3/4	11 1/4	10	5 5/8	4,740	4,740
No. 6	7/8	13 1/2	12	6 3/4	6,730	6,730
No. 7	1	15 3/4	14	7 7/8	9,180	9,180
No. 8	1 1/8	18	16	9	12,090	12,090
No. 9	1 3/8	20 1/4	18	10 1/8	15,150	15,150
No. 10	1 1/2	22 1/2	20	11 1/4	18,770	18,770

For SI: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength. The steel strength values listed are in accordance with Section 2.1.2.3.

²Allowable loads may be increased by 33-1/3% for short-term loading due to earthquakes or wind.

³The tabulated values are for anchors installed with edge distances (c) and spacing (s) in accordance with Table 9. The reduction factors listed in this table shall be applied to the values listed in the bond or concrete strength column.

⁴The tabulated load values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation. Linear interpolation may be used for concrete strengths between those listed.

⁵Allowable loads based on bond strength are calculated based on a safety factor of 4.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

TABLE 12 – SPECIFICATIONS AND DETAILS FOR INSTALLATION OF THREADED ROD IN CONCRETE USING CHEM-STUD ADHESIVE

PROPERTY	ROD DIAMETER (inches)						
	3/8	1/2	5/8	3/4	7/8	1	1-1/4
A _n (Nominal area of rod (in. ²))	0.1105	0.1963	0.3068	0.4418	0.6013	0.7854	1.2272
BD (Nominal bit diameter (in.))	7/16	9/16	11/16	7/8	1	1 1/8	1 3/8
E (Minimum anchor embedment (in.))	3 1/2	4 1/4	5	6 5/8	7	8 1/4	10 1/2
T (Maximum tightening torque (ft.-lbf.))	20	40	90	160	200	300	500

For SI: 1 inch = 25.4 mm, 1 square inch = 645.16 mm², 1 ft.-lbf. = 1.35 N-m

TABLE 13 – MANUFACTURER'S RECOMMENDED CURE TIME FOR CHEM-STUD ADHESIVE

BASE MATERIAL TEMPERATURE ¹ (°F)	FULL CURE TIME ²
0 to 9	24 hours
9 to 14	18 hours
14 to 23	10 hours
23 to 32	5 hours
32 to 50	1 hour
50 to 68	30 minutes
68 and above	20 minutes

For SI: °C = 5/9(°F - 32)

¹This is the temperature as measured in the base material.

²The full curing time is the minimum time required for the adhesive to achieve its ultimate load capacities.

TABLE 14 - REDUCTIONS FOR SPACING AND EDGE DISTANCES FOR THREADED ROD INSTALLED IN CONCRETE WITH CHEM-STUD ADHESIVE

DIMENSION / LOAD TYPE	DISTANCE FOR FULL ANCHOR CAPACITY ¹	MINIMUM DISTANCE ²	REDUCTION FACTOR ³
SPACING (s) BETWEEN ANCHORS – TENSION AND SHEAR LOADS	s _{gr} = 16D	s _{min} = 8D	F _t = F _v = 0.64
EDGE DISTANCE (c) – TENSION LOADS	c _{gr} = 10D	c _{min} = 4D	F _t = 0.64
EDGE DISTANCE (c) – SHEAR LOADS	c _{gr} = 12D	c _{min} = 4D	F _v = 0.22

¹The listed values are the minimum distances required to obtain the load values listed in Table Nos. 14 and 15. D = Anchor Diameter. When adjacent anchors are different sizes or embedments, use largest value for D.

²The listed values are the minimum distances at which the anchor can be set, when load values are adjusted appropriately.

³Load values in the tables are multiplied by the reduction factor when anchors are installed at the minimum spacing listed. Use linear interpolation for spacing between critical and minimum distances. Multiple reduction factors for more than one spacing or edge distance are calculated separately and multiplied.

TABLE 15 - ALLOWABLE TENSION LOADS FOR THREADED ROD INSTALLED IN
NORMAL-WEIGHT CONCRETE WITH CHEM-STUD ADHESIVE 1, 2, 3, 4, 5, 6

ANCHOR DIAMETER (inches)	BIT DIAMETER (inches)	SPACING, s (inches)	EDGE, c (inches)	MINIMUM EMBED. (inches)	ALLOWABLE TENSION LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds)				ALLOWABLE TENSION LOAD BASED ON STEEL STRENGTH (pounds)			
					$f_c=2000$ psi	$f_c=3000$ psi	$f_c=4000$ psi	$f_c=5000$ psi	ASTM A 36	ASTM A 307	ASTM A 193 B7	304 / 316 SS
3/8	7/16	6	3 3/4	3 1/2	1,275	1,885	2,190	2,805	2,115	2,185	4,555	3,630
1/2	9/16	8	5	4 1/4	2,510	2,990	3,390	4,270	3,755	3,885	8,095	6,470
5/8	11/16	10	6 1/4	5	3,460	4,060	4,635	5,850	5,870	6,075	12,665	10,130
3/4	7/8	12	7 1/2	6 5/8	5,575	6,355	7,135	8,695	8,455	8,750	18,225	14,400
7/8	1	14	8 3/4	7	5,870	7,005	8,140	10,410	11,510	11,905	24,805	19,860
1	1 1/8	16	10	8 1/4	8,090	9,885	11,675	13,260	15,035	15,550	32,400	22,020
1 1/4	1 1/2	20	12	10 1/2	14,595	18,225	17,850	20,560	23,485	24,295	50,615	34,420

For SI: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength. The steel strength values listed are in accordance with Section 2.2.2.2.

²Allowable loads may be increased by 33-1/3% for short-term loading due to earthquakes or wind.

³The tabulated values are for anchors installed with edge distance (c) and spacing (s) in accordance with Table 14. The reduction factors listed in this table shall be applied to the values listed in the bond or concrete strength column.

⁴The tabulated load values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation. Linear interpolation may be used for concrete strengths between those listed.

⁵Allowable loads based on bond strength are calculated based on a safety factor of 4.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

TABLE 16 - ALLOWABLE SHEAR LOADS FOR THREADED ROD INSTALLED IN MINIMUM
2000 PSI NORMAL-WEIGHT CONCRETE WITH CHEM-STUD ADHESIVE 1, 2, 3, 4, 5, 6

ANCHOR DIAMETER (inches)	BIT DIAMETER (inches)	SPACING, s (inches)	EDGE, c (inches)	MINIMUM EMBED. (inches)	ALLOWABLE SHEAR LOAD BASED ON STEEL STRENGTH (pounds)			
					ASTM A 36	ASTM A 307	ASTM A 193 B7	304 / 316 SS
3/8	7/16	6	4 1/2	3 1/2	1,090	1,125	2,345	1,870
1/2	9/16	8	6	4 1/4	1,940	2,000	4,170	3,330
5/8	11/16	10	7 1/2	5	3,025	3,130	6,520	5,210
3/4	7/8	12	9	6 5/8	4,355	4,505	9,390	8,390
7/8	1	14	10 1/2	7	5,930	6,135	12,780	8,680
1	1 1/8	16	12	8 1/4	7,745	8,010	16,690	11,340
1 1/4	1 1/2	20	15	10 1/2	12,100	12,915	26,075	17,730

For SI: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength. The steel strength values listed are in accordance with Section 2.2.2.2.

²For short-term loading due to earthquakes or wind allowable shear loads for A 36 & A 307 may be increased by 33-1/3%. A 36 & A 307 values must be used instead of listed values for higher strength steels for these loading conditions.

³The tabulated values are for anchors installed with edge distance (c) and spacing (s) in accordance with Table 14. The reduction factors listed in this table shall be applied to the values listed in the bond or concrete strength column.

⁴The tabulated load values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation. Linear interpolation may be used for concrete strengths between those listed.

⁵Allowable loads based on bond strength are calculated based on a safety factor of 4.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

FIGURE 1 - ALLOWABLE TENSION LOAD BOND STRENGTH REDUCTION BASED ON IN-SERVICE TEMPERATURE FOR THE POWER-FAST EPOXYADHESIVE

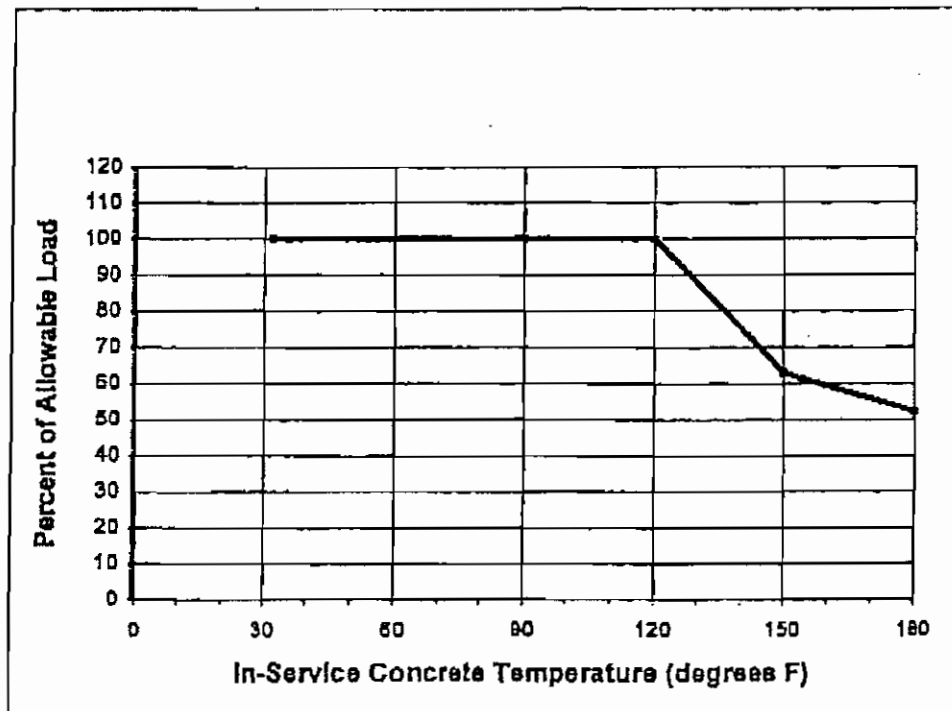
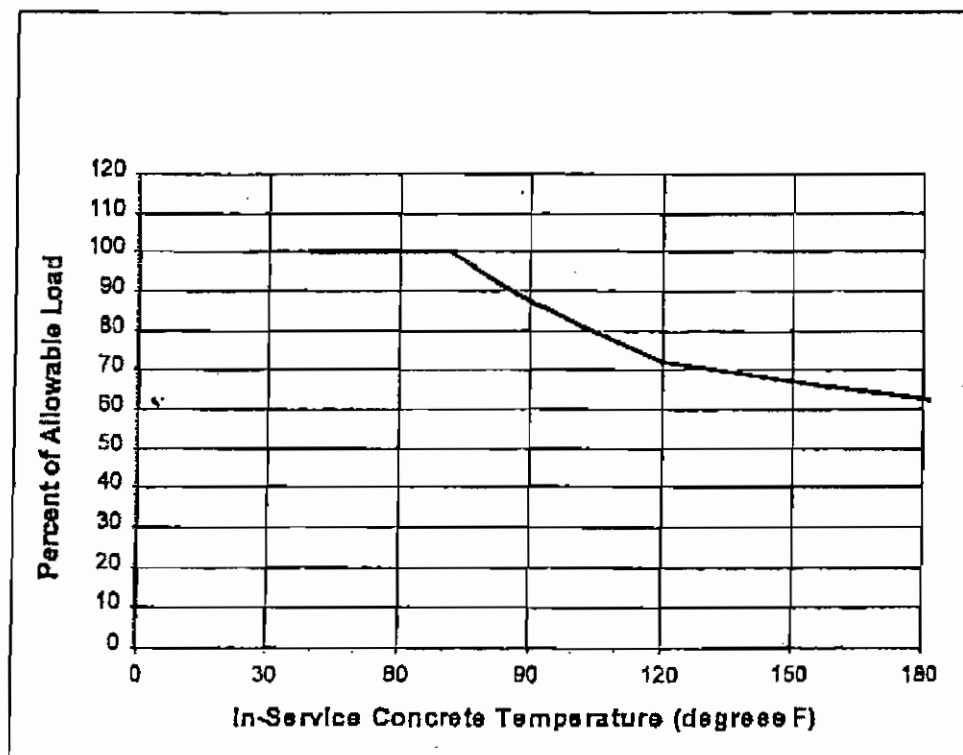


FIGURE 2 - ALLOWABLE TENSION LOAD BOND STRENGTH REDUCTION BASED ON IN-SERVICE TEMPERATURE FOR THE CHEM-STUD ADHESIVE



Product Profile

Power-Fast™ System Provides Quick, Easy, Reliable Anchorage In New Suspended Ceiling for Boston's Callahan Tunnel

To secure the anchors for the ceiling hangers supporting a new suspended ceiling in Boston's Callahan Tunnel, the Power-Fast™ Epoxy Injection Gel Anchoring System was selected because it was less costly than the encapsulated alternative; was odor free, an important advantage in the enclosed work area; and did not require any special training of the work force.

Originally, the ceiling of the approximately one-mile long tunnel — a major outbound artery from downtown Boston — was a 6" thick reinforced concrete that had aged considerably in its ear service life. Ceramic tiles on the face of the ceiling were falling off onto the roadway and represented a potential hazard to passing motorists. In addition, the time and manpower needed to remove the debris from the road was costly. Replacement of the fallen tiles was not feasible because proper adhesion of tile to concrete could not be assured. Numerous alternatives to remedy this hazard were

thoroughly investigated, including the possibility of patching identified areas of deteriorated concrete throughout the entire length of tunnel.

The operating agency of the tunnel, the Massachusetts Turnpike Authority, determined that the most effective way to resolve the issue was to remove and replace the ceiling.

The design selected was a suspended ceiling comprised of 2" thick porcelain enameled steel face sheets that were bonded to a fire resistant material. The panels were mounted onto structural steel stringers suspended from hangers that were secured by anchors installed into the overhead sections of the concrete facing of the structural steel tunnel lining.

"We specified an adhesive anchoring system rather than mechanical anchors," stated a spokesperson for HNTB, Boston, the design firm for the ceiling project. "The concrete in the tunnel is more than a quarter century old and is subject to constant exposure to moisture. The variations that exist in the concrete due to its age and its moisture content have a direct bearing on the ability of a mechanical anchor to produce a proper hold in every instance," he explained.

"Also," he added, "we know that if an anchor did not hold, special measures would have to be taken to correct the problem. If this occurred repeatedly, which we expected, it would cause serious time delays that were totally unacceptable because of this project's rigid completion schedule of one year.

Product:

Power-Fast™ Epoxy Injection Gel

Application:

Secure anchors for ceiling hangers

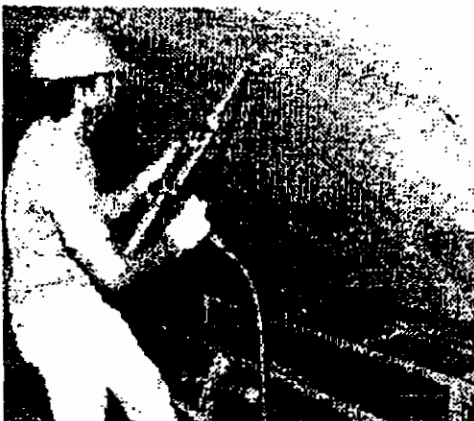
Location:

**Callahan Tunnel
Boston, Mass.**

"Based on previous experience with adhesive anchors in the adjoining Sumner Tunnel, we knew that a moisture and vibration resistant adhesive anchor would be better suited for this application," he continued. "It would be less dependent upon the condition of the concrete or the degree of moisture absorption to produce and maintain a suitable fastening. Similarly, vibration generated by the traffic below would not affect the integrity of the connection. Also, adhesive anchors can be installed quickly and easily, good characteristics for jobs with tight time constraints."

The subcontractor on the project, Webster Engineering, Dorchester, MA, selected the Power-Fast Epoxy Injection Gel System. This high strength, comprehensive system from Powers Fasteners, Inc., consists of Sikadur® Injection Gel, a two-component structural epoxy that is dispensed from either manually or pneumatically operated tools for accurate mixing.

(continued on other side)



For additional information, contact your local Powers branch or Powers Fasteners, Inc.
2 Powers Square, New Rochelle, NY 10801
(914) 235-6300 • www.powers.com

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No.PF-003

The gel is an amine-based material that has a very high bond strength. In addition to providing a high strength anchorage, it produces no foul odor and offers excellent resistance to moisture and chemicals. The adhesive meets the requirements of ASTM C-881.

Prior to its selection, the Power-Fast system was subjected to a series of pullout tests. The Powers distributor, Boston Cartridge Tool Co., set up these tests which were witnessed by an Inspector from the Massachusetts Turnpike Authority. Based on a design load of 4,000 pounds per anchor, the tests were run at four times the design load. Actual pullout values averaged six times the design load, which were attributed to the strength of the adhesive.

The failure rating of the adhesive was never determined because the

concrete failed long before the adhesive showed any loss of holding power.

"Although this is the first time we have worked with epoxy on such a large scale project, this system was so easy to use that our work crews required



only one day to acclimate themselves with its operation," commented Paul Wilkins, General Superintendent of Webster. "Boston Cartridge Tool was extremely helpful in providing the instruction our crews needed to quickly and efficiently use the system.

"For example," he added, "for the ceiling hangers, we had to install over 3,000 3/4" diameter stainless steel anchors, spaced every five feet across the width of the tunnel. In addition, we had to install over 1,800 1/2" diameter stainless steel anchors, two every five feet, to secure the utility tray. For the 3/4" diameter anchor, a two-man crew installed about 150 units a day. The same size work force installed the 1/2" diameter anchors at a rate of 200 per day, thereby keeping us within our target completion date," he concluded.